Model diagnostics

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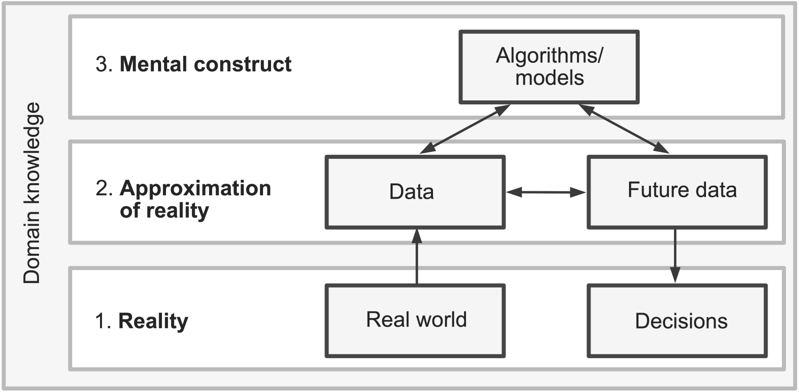
Model assessment

Essentially, all models are wrong, but some are useful. (Box and Draper, 1987)



Question: Is our model useful?

(Image from *Veridical Data Science*, Yu and Barter)

Model assessment

We observe data $(\mathbf{x}_1, Y_1), ..., (\mathbf{x}_n, Y_n)$ and fit the model

$$Y_i | \mathbf{x}_i \sim Bernoulli(p_i)$$
$$\log\left(\frac{p_i}{1 - p_i}\right) = \mathbf{x}_i^T \beta$$

How do we assess whether the model is useful?

Model assumptions

$$Y_i|\mathbf{x}_i \sim Bernoulli(p_i)$$
$$\log\left(\frac{p_i}{1-p_i}\right) = \mathbf{x}_i^T\beta$$

What assumptions do we make when fitting this logistic regression model?

Shape assumption

$$Y_i | \mathbf{x}_i \sim Bernoulli(p_i)$$
$$\log\left(\frac{p_i}{1-p_i}\right) = \mathbf{x}_i^T \beta$$

Shape assumption: The log-odds are a linear function of the explanatory variables (i.e., the systematic component is correctly specified).

Question: How do we assess the shape assumption in a *linear* regression setting?

Residual plot?

Raw residuals: $Y_i - \widehat{p}_i$

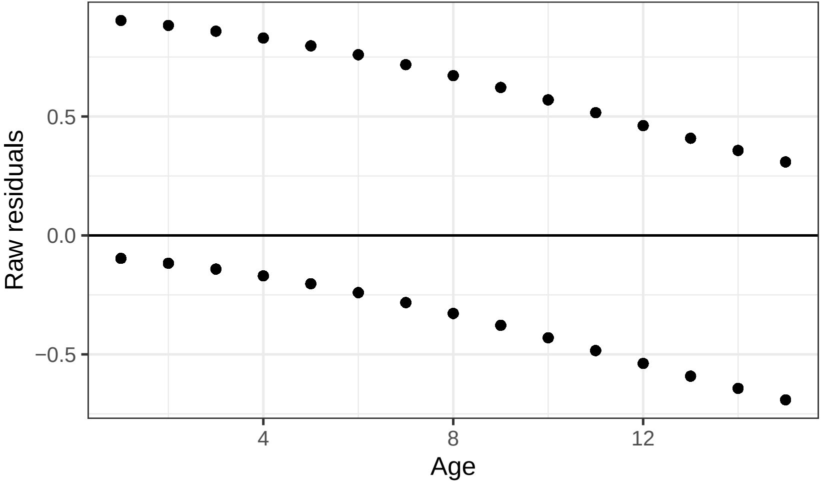
Question: what will the plot look like if we plot $Y - \widehat{p}$ vs. X?

Residual plot?

Raw residuals: $Y_i - \widehat{p}_i$

Question: what will the plot look like if we plot $Y - \widehat{p}$ vs. X?

Example: predicting dengue status with Age

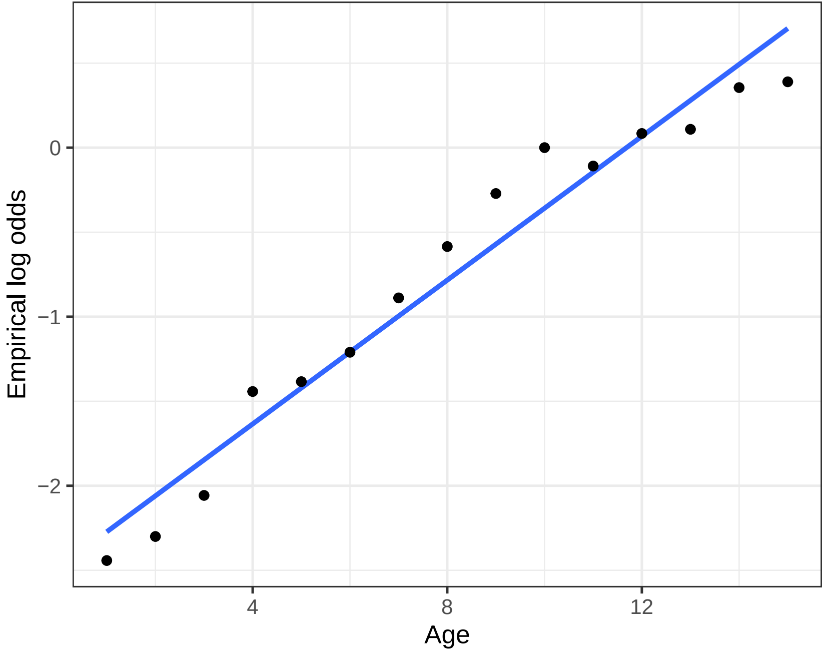


Alternative: empirical logits

For each Age, calculate the *empirical* log odds of dengue (aka *empirical logits*):

```
## # A tibble: 15 x 5
##      Age NoDengue YesDengue      p log_odds
##    <int>    <int>     <int>  <dbl>    <dbl>
##  1     1      184        16 0.08     -2.44
##  2     2      419        42 0.0911   -2.30
##  3     3      532        68 0.113    -2.06
##  4     4      457       108 0.191    -1.44
##  5     5      487       122 0.200    -1.38
##  6     6      369       110 0.230    -1.21
##  7     7      360       148 0.291    -0.889
##  8     8      289       161 0.358    -0.585
##  9     9      244       186 0.433    -0.271
## 10    10      185       185 0.5       0
## 11    11      165       148 0.473    -0.109
## 12    12      126       137 0.521     0.0837
## 13    13       96       107 0.527     0.108
```

Empirical logit plot



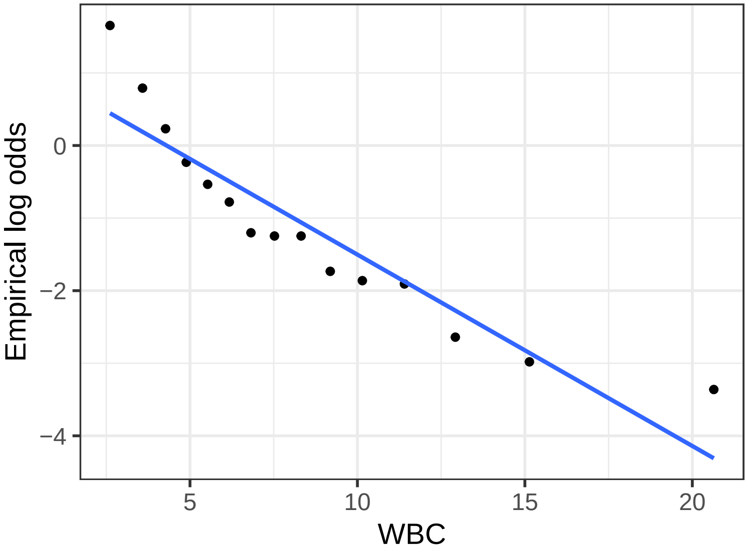
Empirical logits with a continuous variable

Trying with WBC:

```
## # A tibble: 6 x 5
##     WBC NoDengue YesDengue     p log_odds
##   <dbl>    <int>     <int> <dbl>    <dbl>
## 1  0.9         0         1     1      Inf
## 2  1.15        0         2     1      Inf
## 3  1.23        0         1     1      Inf
## 4  1.25        0         1     1      Inf
## 5  1.54        0         3     1      Inf
## 6  1.58        0         1     1      Inf
```

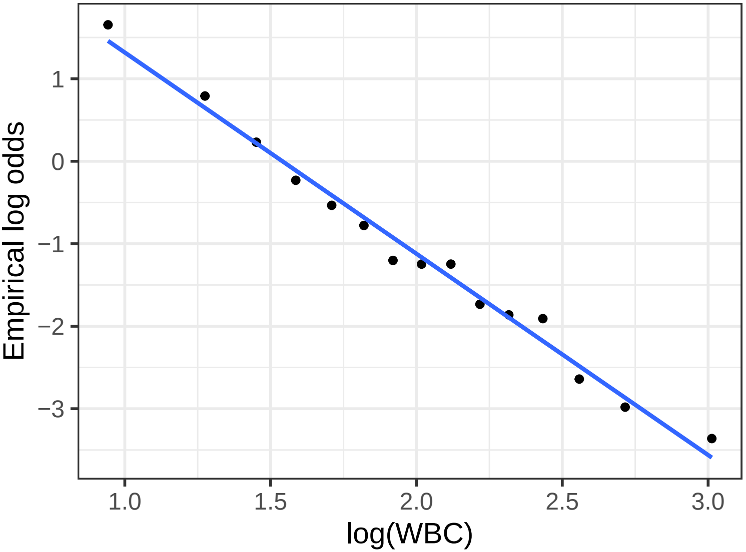
What has gone wrong? What should we do differently?

Binned empirical logit plots

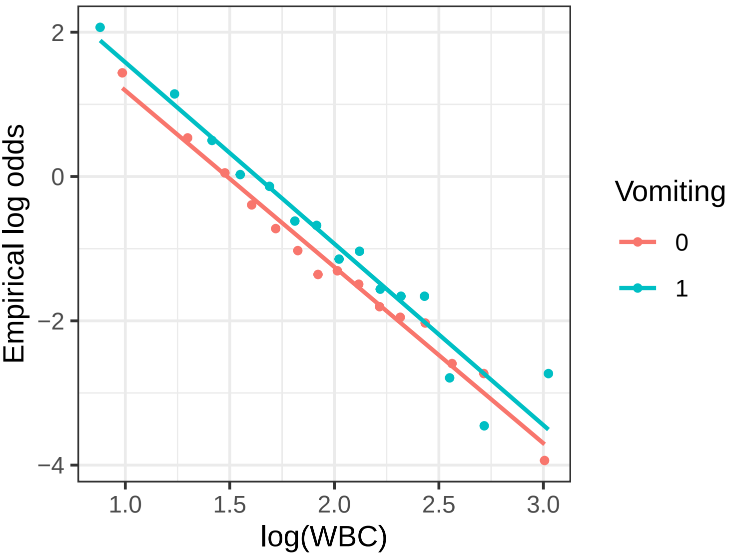


Does the shape assumption seem reasonable?

Binned empirical logit plots



Binned empirical logit plots



Limitations of empirical logit plots

- Does not assess a fitted model (performed *before* fitting the model)
- One point per *bin*, not one point per *observation*
- Can only plot one or two explanatory variables at a time

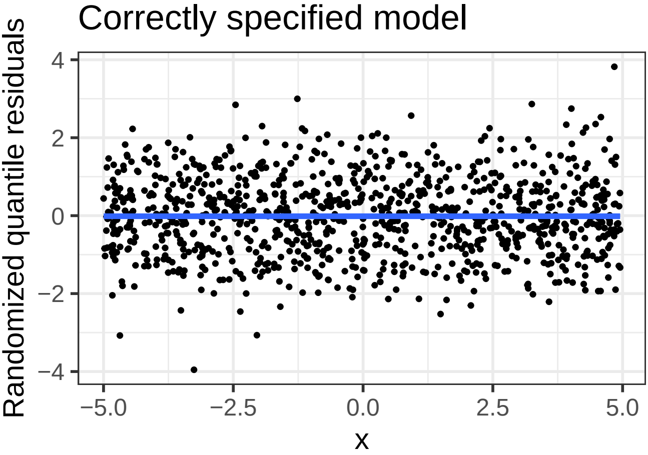
Randomized quantile residuals

Define *randomized quantile residuals* $r_{Q,i}$:

Randomized quantile residuals

Simulated data: $\log\left(\frac{p_i}{1-p_i}\right) = -1 + 2X_i$

Model fit: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i$



Randomized quantile residuals

Simulated data: $\log\left(\frac{p_i}{1-p_i}\right) = -1 + 2X_i^2$

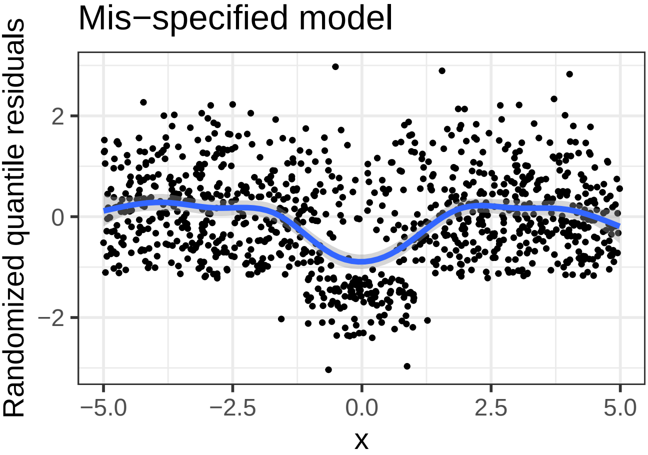
Model fit: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i$

Question: What do you think the randomized quantile residual plot will look like?

Randomized quantile residuals

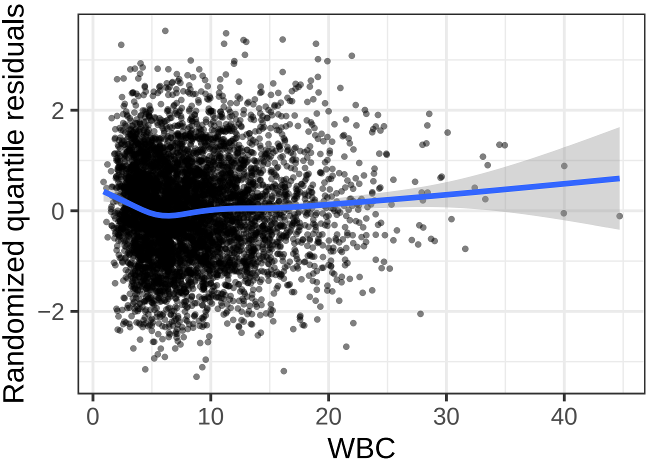
Simulated data: $\log\left(\frac{p_i}{1-p_i}\right) = -1 + 2X_i^2$

Model fit: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i$



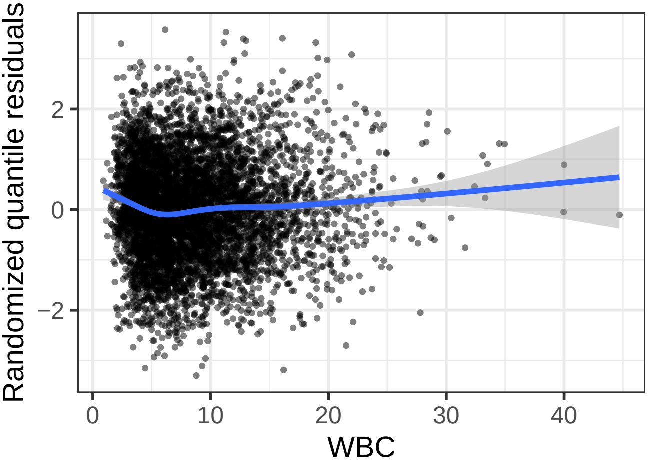
Example: dengue data

Model: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{WBC}_i$



Example: dengue data

Model: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{WBC}_i$



Question: How can we decide whether a plot is consistent with a correctly-specified model?

Parametric bootstrap

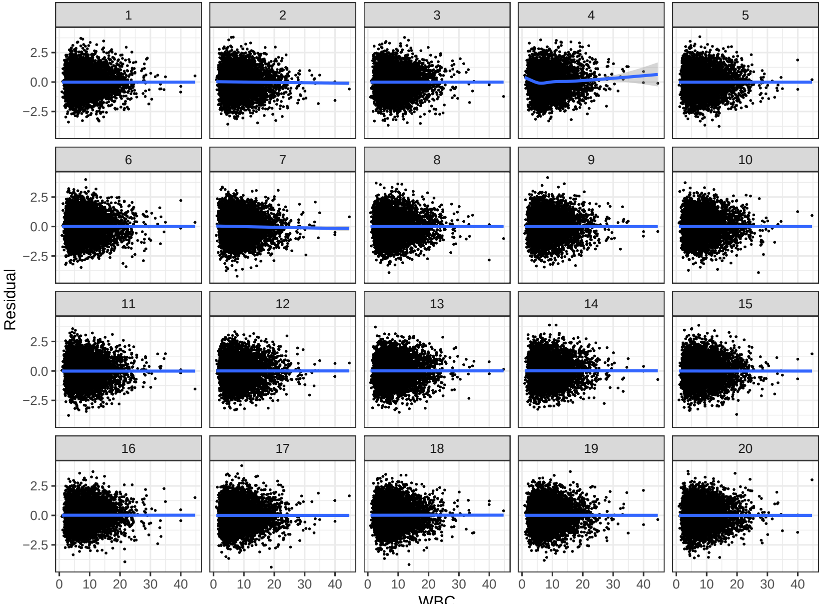
The *parametric bootstrap* generates data from the *fitted* model:

Class activity

Work on the class activity (available on the course website).

Visual inference

```
## decrypt("QUg2 qFyF Rx 8tLRyRtx Ko")
```



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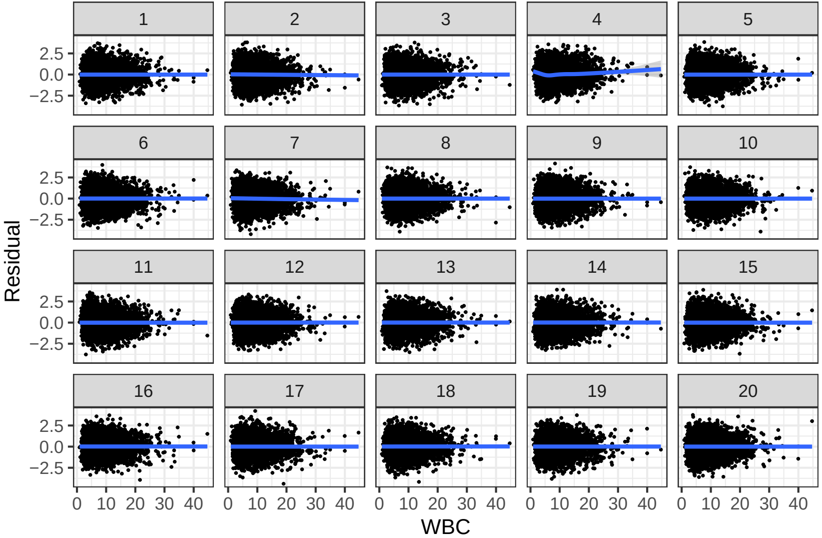


```r
decrypt("QUg2 qFyF Rx 8tLRyRtx Ko")
```

```
## [1] "True data in position  4"
```